

Shift Factors on the Irish Transmission Grid

Constraint groups, computed analytically

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1. A new shift factor, $S_{e,n}$

The change in flow on branch e per MW curtailed at wind node n . Positive means relief, negative means it loads the branch further.

$$\text{PTDF} = BK^T L^+, \quad S_{e,:} = \text{PTDF}_{e,:} \cdot \text{sign}(f_e)$$

A 980×754 matrix, built once from topology and reactance — 6 branches reach their rating over the week, 0 of them transformers.

2. Why it's better

- ▶ **EirGrid:** perturb one farm by 10 MW, re-run the load flow, read the change. One number per run.
- ▶ **This method:** an analytic derivative — all 980×157 branch–node pairs at once, exact to the DC model.
- ▶ **Verified** against an independent solver (PyPSA): residual = $2.1e - 09$ MW.

3. The issue with SF constraint groups, and the fix

EirGrid's group: $\{n : S_{e,n} \geq \tau\}$, $\tau = 0.05$ — one row, read in isolation. Blind to every other branch.

Net sensitivity alone made it **worse**: 1,085 events broke a line, against 642 for the plain method.

The veto: cap each node's cut at the point where any other live branch would be pushed over its rating.

	cleared	broke a line
Shift-factor threshold (today)	92.0%	642 events
Net sensitivity + veto	89.9%	0 events

20,203 (outage, hour) events.

4. Fixed for every N-1 contingency, computed once

- ▶ Group membership depends only on topology and reactance — not hour, not weather.
- ▶ Precomputed for all 432 outages → 58 circuits ever at risk, 1,308 lookup rows.
- ▶ 10.0% of EirGrid's own group memberships fail the veto and are excluded before dispatch.
- ▶ Online, only the live flow and the final MW instruction are computed.

5. Farms that can run full blast

- ▶ Top 5 nodes account for 46% of all MW ever curtailed; top 20 for 72%.
- ▶ 119 of 157 nodes — 65.9% of fleet capacity — are cut in under 1% of events.
- ▶ Across the entire N-1 study, only 0.85% of all wind offered is ever curtailed.

6. Application: stopping cascading outages

250 starting outages, chain run to exhaustion:

	lines lost	spread	islanded
No remedy	814	250 / 250	169
EirGrid threshold	309	137 / 250	72
Net sensitivity + veto	261	119 / 250	55
Safe optimum (LP)	38	25 / 250	7

Live demo: [cascade3.html](#)